Proof. Let V be irreducible. For any element $a \in A$, the operator $\rho(a) : V \to V$ is an intertwining operator. Indeed,

$$\rho(a)\rho(b)v = \rho(ab)v = \rho(ba)v = \rho(b)\rho(a)v$$

(the second equality is true since the algebra is commutative). Thus, by Schur's lemma, $\rho(a)$ is a scalar operator for any $a \in A$. Hence every subspace of V is a subrepresentation. But V is irreducible, so 0 and V are the only subspaces of V. This means that $\dim V = 1$ (since $V \neq 0$). $\square$

Example 2.3.14. 1. $A = k$. Since representations of A are simply vector spaces, $V = A$ is the only irreducible and the only indecomposable representation.

2. $A = k[x]$. Since this algebra is commutative, the irreducible finite dimensional representations of A are its 1-dimensional representations. As we discussed above, they are defined by a single operator $\rho(x)$. In the 1-dimensional case, this is just a number from k. So all the irreducible finite dimensional representations of A are $V_\lambda = k$, $\lambda \in k$, in which the action of A is defined by $\rho(x) = \lambda$. Clearly, these representations are pairwise nonisomorphic.

The classification of finite dimensional indecomposable representations of $k[x]$ is more interesting. To obtain it, recall that any linear operator on a finite dimensional vector space V can be brought to Jordan normal form. More specifically, recall that the Jordan block $J_{\lambda,n}$ is the operator on k^n which in the standard basis is given by the formulas $J_{\lambda,n}e_i = \lambda e_i + e_{i-1}$ for $i > 1$ and $J_{\lambda,n}e_1 = \lambda e_1$. Then for any linear operator $B : V \to V$ there exists a basis of V such that the matrix of B in this basis is a direct sum of Jordan blocks. This implies that all the indecomposable finite dimensional representations of A are $V_{\lambda,n} = k^n$, $\lambda \in k$, with $\rho(x) = J_{\lambda,n}$. The fact that these representations are indecomposable and pairwise nonisomorphic follows from the Jordan normal form theorem (which in particular says that the Jordan normal form of an operator is unique up to permutation of blocks).

This example shows that an indecomposable representation of an algebra need not be irreducible.